



Mangrove Journal **2019-2020**



**Sustainable Forests & Livelihoods
(SUFAL) Project**

**“Mangrove Plantation”
200.0 Hectare**

**Range : charbata Range
Beat : Bashkhali Beat**

**Seedlings : 8,88,800 Nos.
(For 200.0 Hectare Plantation)**

Coastal Forest Division, Noakhali.

Raising For Mangrove 200.0 Hectare Plantation in Financial Year 2019-2020 under SUFAL Project.

Basic Information:

- | | | |
|---------------------|---|--|
| 1. District | : | Noakhali |
| 2. Forest Division | : | Coastal Forest Division, Noakhali. |
| 3. Upazila | : | Subarnachar |
| 4. Range | : | Charbata Range |
| 5. Beat | : | Bashkhali Beat |
| 6. Union | : | Charbata |
| 7. Project | : | Sustainable Forests & Livelihoods (SUFAL) Project. |
| 8. Species | : | Mangrove species "Keora" (<u>Sonneratia apetala</u>) |
| 9. No. of seedlings | : | 8,88,800 Nos. |

Characteristic of the Plantation Site:-		
1.	ArW... of the Plantation	200.0 hac.Mangrove Plantation in New Char Nur Islam of the Year of 2019-20.
2.	Edaphic Condition	Physiographically the Char is Part of the young Meghna Estuarine flood plain.Hobgoblin, the site is soft and clayey and it is also comparatively compact,harder and at the same time subjected to more saline condition.These soils permit easy penetration of mangrove tree roots because of being quite fertile along with soil conditions.
3.	Climate Condition	The New Char Nur Islam has a tropical monsoon climate,characterized by basically four seasons per year.The average temperature usually ranges between 11° to 29°C in winter and 21° to 38°C during summer.The average rainfall is 3257 mm (Islam 2009) and varies from 25000 mm to 4500 mm.Cyclones pass regularly,particularly at the beginning and at the end of the monsoon. The Char is slightly raised flat land which generally inundates twice a day noticeably from March to December.The rate of site deposition is higher particularly from November to February roughly 3"-4".
4.	Selection Criteria of The Planting Site	The Plantation site has been selected through the following criteria. Land suitability: The New Char Nur Islam is newly accreted mud flats (char land).This site is inundated at each high tide,twice per day.This site has lot of fine mud on the surface that makes walking difficult with up to knee deep mud.This site starts bearing dhanshi grasses.This area that has a thicker clayey layer and lower salinity re most suitable for planting with Keora(Sonneratia apetala).One should be able to walk on this site with less difficulty than the potential Keora planting sites.The purpose of mangrove plantations is for protection to improve the resilience of coastal communities and their livelihoods.The mangrove plantations can't be felled and retained for protection from tidal swell and cyclonic. Grazing Problem: It is known to there entire are a number of chars in Noakhali and accreting new lands.These chars are threat to the grazing problems But this is not free from such unavoidable phenomenon.That is why this site selected for the mangrove plantation.

Name & Designation of the Officers, stuffs Involved in the Raising Mangrove Plantation Activities

- 1.Mr. Bipul Krishna Das, DCF.
- 2.Mr. Md. Firoj Alom Chowdhury, ACF (A.C)
- 3.Mr. Abdul Wahab Sheikh, Forester, Range Officer.
- 4.Mr. Md. Sadek Ali, Forester, Beat Officer.
- 5.Mr. Md. Anowar Hossain, Fg.
- 6.Mr. Md. Lokman Hakem, Fg.
- 7.Mr. Md. SanaUllah, Fg..
- 8.Mr. Ripon Miah, Fg..
- 9.Mr. Md. Abdur Rahim , Bm.
- 10.Mr. Md. Momin Ullah, Bm.
- 11.Mr. Md. Akbor Hossain , Bm.
- 12.Mr. Md. Nur Hossain, Pm.
- 13.Mr. Md.Saifuddin Manik, Pm.
- 14.Mr. Md. Masudur Rahaman, Pm.
- 15.Mr. Md. Azad Hossain, Pm.
- 16.Mr. Md. Abul Kasham, Loskar

1. Signature of the officers Involved in the Raising Mangrove Platation Activities :

Sl.	Name	Designation	Signature
01	Mr. Md. Sadek Ali, Forester	Beat Officer, Bashkhali Beat Charbata Range, Coastal Forest Division Noakhali.	 Md. Sadek Ali, Forester Beat Officer, Bashkhali Beat Charbata Range, Coastal Forest Division, Noakhali.
02	Mr. Abdul Wahab Sheikh, Forester	Range Officer Charbata Range, Coastal Forest Division Noakhali.	 Abdul Wahab Sheikh Range Officer Charbata Range, Coastal Forest Division, Noakhali.
03	Mr. Md. Firoj Alom Chowdhury, Forest Ranger	Assistant Conservator of Forest (A.C) Coastal Forest Division Noakhali.	 Md. Firoj Alom Chowdhury Assistant Conservator of Forest (A.C) Coastal Forest Division Noakhali.
04	Mr. Bipul Krishna Das, DCF.	Divisional Forest Officer Coastal Forest Division Noakhali.	 Bipul Krishna Das Divisional Forest Officer Coastal Forest Division Noakhali.

5.	Climate Resilient Aspect of the Plantation	The Coastal forest,breakwater,seawall is to absorb the impact forces and to retard the flow of large storm waves and cyclone.A seawall,if tall enough reflects the waves back out to sea.On the other hand,permeable structures, like breakwaters and coastal forests partly reflect and partly transmit the water.In the case of a coastal forest,energy is progressively absorbed as it passes through the forest.Without the forest barrier,the tsunami/cycone will run-up to a maximum height determined by the magnitude and nature of the seismic event that created the tsunami and local factors such as the coastal profile,offshore bathymetry and beach slope that modify the wave's force.Once the tsunami comes on shore,the amount of reduction in water depth,velocity and force depends on how much water is reflected and energy is absorbed by the coastal forest. Through afforestation mangrove based seedlings biodiversity conservation,human security and so on will be ensured in the project area.Therefore,greenbelt of the project area by raising said plantation has been planned to brig back the balance of ecosystem into the same char lands and to develop a friendly environment by raising coastal afforestation.It is for thfact that the land is sustainable for mangrove and non-mangrove planttions where bio-physical and socio-economic conditions will improve.The new Char Nur Islam will be brought to work as a barrier during natural disasters by dint of the mangrove Plantation.
	Other Activities	
1.	Delineation of the Planting Area	The New Char Nur Islam is a newly formed char land which has raised rapidly with in a short time.The char is adjacent to the East-North side of Keora plantation.The land initially devoid any vegetation except dens Urigrass as first colonizer.The soils of the side permit easy penetration of mangrove tree roots and agriculturl crop roots because of being fertile.The project site would be used for different types of species diversity plantation.If the coastal forest lands of a target area are properly replenished with greens,the existing wildlife habitats will increase and will be a nice environment for biodiversity conservation.This type of coastal forest lands can fight back the prevailing environmental threats and mitigate the negative inpacts of the climate changes took place in these coastal zones.Thus,this type of coastal forest lands not only save the life and resources from natural disasters but also play important role in economic development.
2.	GPS Reading of all Corners of the Planting Site	The GPS has been taken from all corners of the plantation site in presence of the RO.BO,BM as will as all other staff officer of the Charbata Range,Coastal Forest Divition,Noakhali.
3.	Post-Planting Survey,Surveyor,Survey Technique & Record	Md. Shafiqul Islam Surveyor The Survey has been completed through the technique of GPS reading and it is fistly recorded in the note book after that the record is shown in the map.

EXPENDITURE SHEET

Range : charbata Range

Project : Sustainable Forests & Livelihoods (SUFAL) Project

Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
1/chaba	January /2020	Survey the plantation site and prepare plantation site map and Journal Raising for 100.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY.all complete by Daily labours 100 Nos. @500/-	50,000/-	50,000/-	25/588
2/chaba	January /2020	Uprooting of 4,44,400 seedlings at the nursery site strip of the lower leaves leaving 3 to 4 top leaves. prune the excessive lateral roots. Place them systematically on wet gunny bags or hogla mat. For loading on the boat for transportation Raising for 100.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. all complete by by Daily labours 200 Nos. @500/- .	1,00,000/-	1,50,000/-	26/588
3/chaba	January /2020	Boat hire for Transport the seedlings to plantation site Raising for 100.0 hac. Out of 200.0 hac. Mangrove Plantation	25,000/-	1,75,000/-	27/588
4/chaba	January /2020	Do	25,000/-	2,00,000/-	28/588
5/chaba	January /2020	Do	25,000/-	2,25,000/-	29/588
6/chaba	January /2020	Do	25,000/-	2,50,000/-	30/588
7/chaba	January /2020	Do	25,000/-	2,75,000/-	31/588
8/chaba	January /2020	Do	25,000/-	3,00,000/-	32/588
9/chaba	January /2020	Do	25,000/-	3,25,000/-	33/588
10/chab a	January /2020	Do	25,000/-	3,50,000/-	34/588
11/chab a	January /2020	Do	20,000/-	3,70,000/-	35/588



Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
12/chaba	January/2020	Transport the seedlings to plantation site mostly by boat including loading and unloading Raising for 100.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labours 100 Nos. @500/-	50,000/-	4,20,000/-	36/588
13/chaba	January/2020	Cleaning of dhanshi in lines and planting of the bare rooted seedlings carefully at 1.5 m X 1.5 m spacing Raising for 100.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labors 500 Nos. @500/-	2,50,000/-	6,70,000/-	37/588
14/chaba	January/2020	Cleaning of dhanshi in lines and planting of the bare rooted seedlings carefully at 1.5 m X 1.5 m spacing Raising for 100.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labors 150 Nos. @500/-	75,000/-	7,45,000/-	38/588
01/chaba	February/2020	Uprooting of 4,44,400 seedlings at the nursery site strip of the lower leaves leaving 3 to 4 top leaves. prune the excessive lateral roots. Place them systematically on wet gunny bags or hogla mat. For loading on the boat for transportation Raising for Final payment 200.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labors 200 Nos. @500/-	1,00,000/-	8,45,000/-	40/588
02/chaba	February/2020	Boat hire for Transport the seedlings to plantation site Raising for 100.0 hac. Out of 200.0 hac. Mangrove Plantation in 2019-20 FY.	24,000/-	8,69,000/-	41/588
03/chaba	February/2020	Do	24,000/-	8,93,000/-	42/588
04/chaba	February/2020	Do	24,000/-	9,17,000/-	43/588
05/chaba	February/2020	Do	24,000/-	9,41,000/-	44/588
06/chaba	February/2020	Do	24,000/-	9,65,000/-	45/588
07/chaba	February/2020	Do	24,000/-	9,89,000/-	46/588
08/chaba	February/2020	Do	24,000/-	10,13,000/-	47/588
09/chaba	February/2020	Do	24,000/-	10,37,000/-	48/588
10/chaba	February/2020	Do	24,000/-	10,61,000/-	49/588
11/chaba	February/2020	Do	4,000/-	10,65,000/-	50/588
12/chaba	February/2020	Transport the seedlings to plantation site mostly by boat including loading and unloading Raising for Final payment 200.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labours 100 Nos. @500/-	50,000/-	11,15,000/-	51/588

Vr. No.	Month	Particulars	Amount		NBP
			Current	Progress	
13/chaba	February/ 2020	Cleaning of dhanshi in lines and planting of the bare rooted seedlings carefully at 1.5 m X 1.5 m spacing Raising for Final payment 200.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labours 500 Nos. @500/-	2,50,000/-	13,65,000/-	52/588
14/chaba	February/ 2020	Cleaning of dhanshi in lines and planting of the bare rooted seedlings carefully at 1.5 m X 1.5 m spacing Raising for Final payment 200.0 hac. out of 200.0 hac. mangrove Plantation in 2019-20 FY. Daily labors 150 Nos. @500/-	75,000/-	14,40,000/-	53/588
		Cotation	70,000/-	15,10,000/-	


 Abdul Wahab Sheikh
 Range Officer
 Charbata Range

Sustainable Forests & Livelihoods (SUFAL) Project.

Coastal Forest Division, Noakhali

Map of Mangrove Plantation, Range : Charbata Range

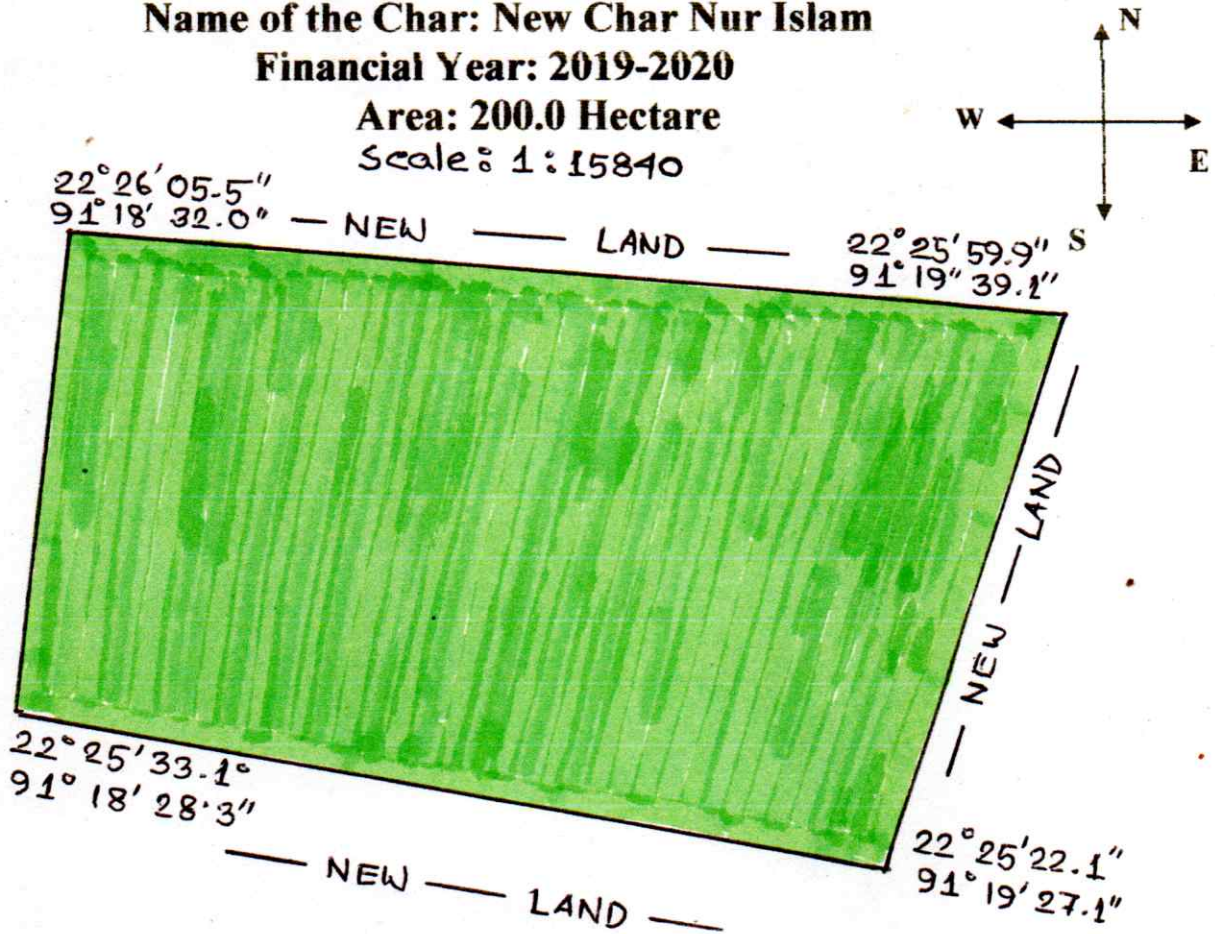
Beat : Bashkhali Beat

Name of the Char: New Char Nur Islam

Financial Year: 2019-2020

Area: 200.0 Hectare

Scale: 1:15840



Symbols:-

• Plantation Area



• Khal



Countersigned

Abdul Wahab Sheikh
Range Officer
Charbata Range

Submitted By

Md. Sadek Ali, Forester
Beat Officer, Bashkhali Beat
Charbata Range
Coastal Forest Division, Noakhali

Prepared By

Md. Shauqul Islam
Surveyor
Coastal Forest Division, Noakhali

মোঃ ফিরোজ আলম চৌধুরী
প্রকল্প পরিচালক (নোয়াখালী জেলা)
উপকূলীয় বন বিভাগ, নোয়াখালী।

Remarks of Inspection Officer

Date	Description	Remarks